

LAB EXPERIMENTS(11-15)

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Set 11:

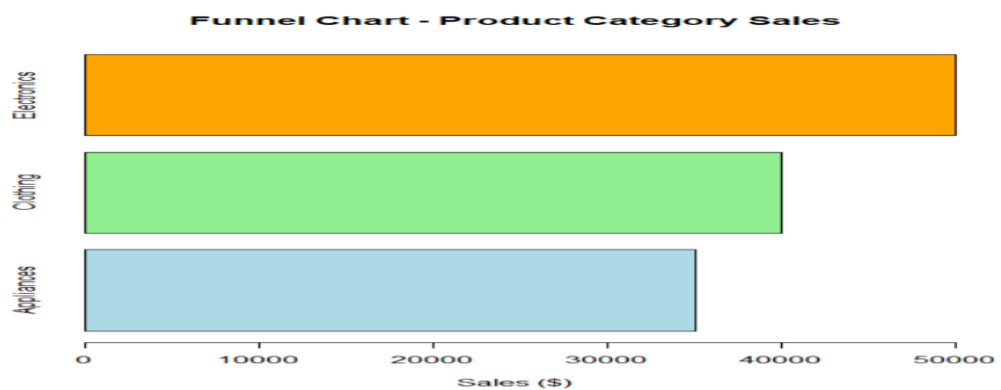
Product Category Analysis

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

CODE:

```
# Dataset
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 40000, 35000)
# Funnel-style chart
par(mar = c(5,8,4,2))
barplot(
  rev(sales),
  horiz = TRUE,
  names.arg = rev(category),
  col = c("lightblue", "lightgreen", "orange"),
  xlab = "Sales ($)",
  main = "Funnel Chart - Product Category Sales")
```

OUTPUT:



2. Build a table to display the sales data for each product category. Label the table elements.

CODE:

```
# Dataset
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)

# Create Table
```

```
product_table <- data.frame(
  Category = category,
  Sales = sales
)

print(product_table)
```

OUTPUT:

```
print(product_table)
  Category Sales
Electronics 50000
  Clothing 35000
  Appliances 40000
```

3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

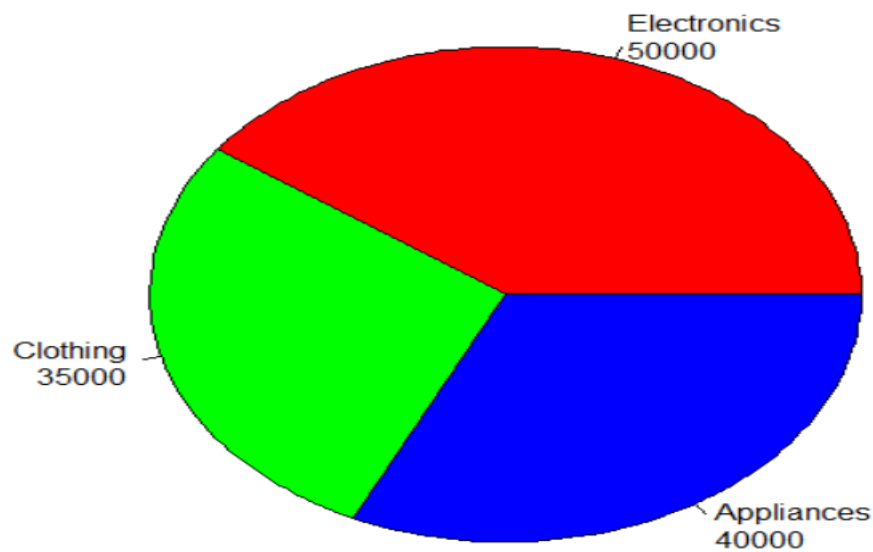
CODE:

```
# Dataset
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)

# Pie Chart
pie(
  sales,
  labels = paste(category, sales, sep = "\n"),
  col = c("red", "green", "blue"),
  main = "Sales Distribution Across Product Categories"
)
```

OUTPUT:

Sales Distribution Across Product Categories



Set 12 :

Dataset: Website Traffic

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

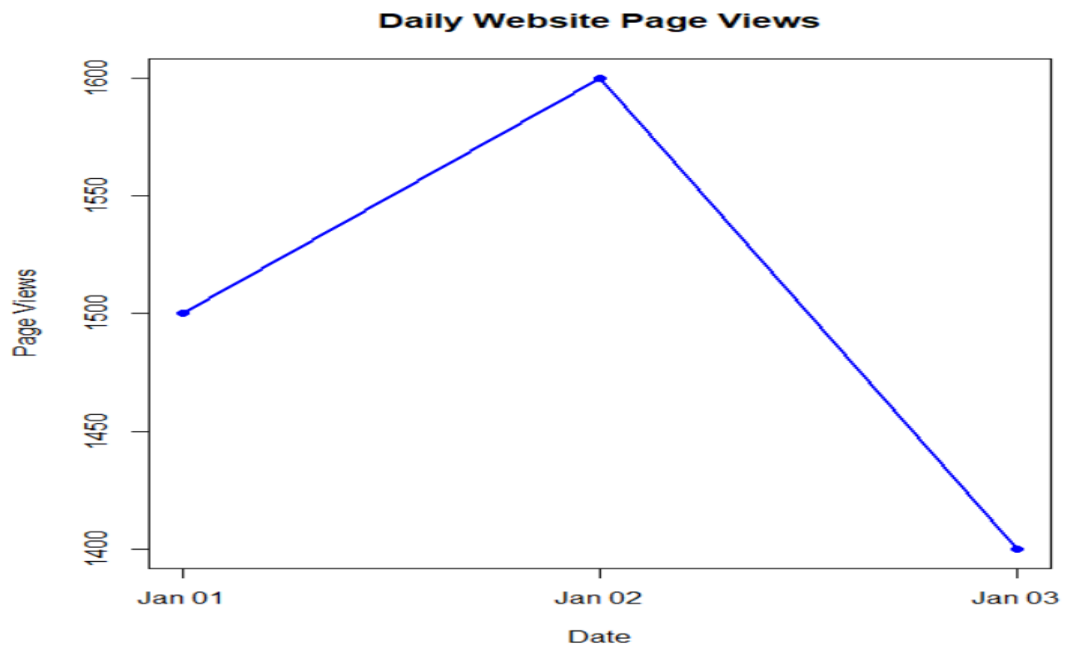
CODE:

```
# Dataset

date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
page_views <- c(1500, 1600, 1400)

# Line Chart
plot(date,
      page_views,
      type="o",
      col="blue",
      pch=16,
      lwd=2,
      xlab="Date",
      ylab="Page Views",
      main="Daily Website Page Views")
```

OUTPUT:

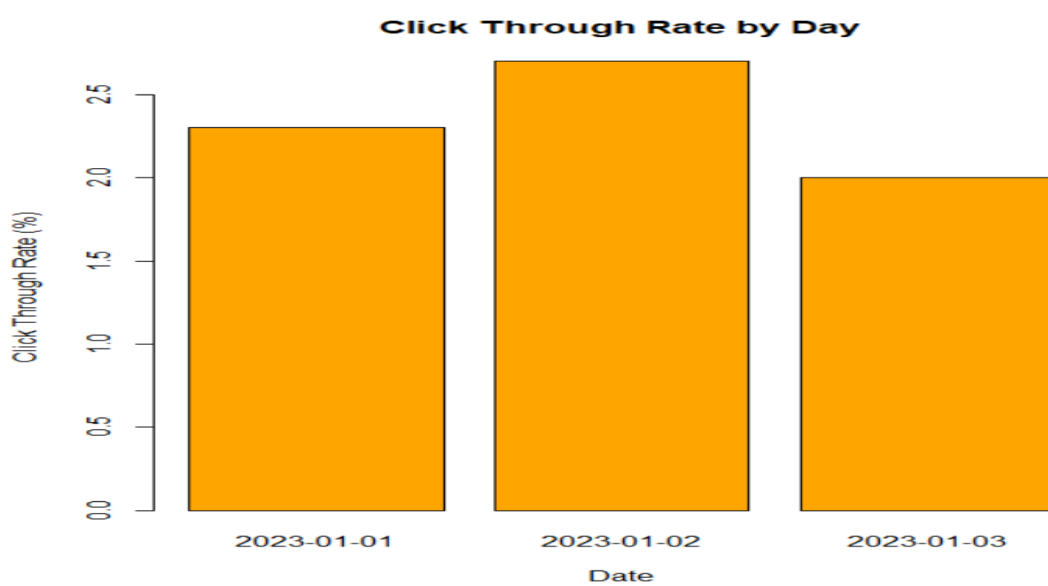


2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

CODE:

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
ctr <- c(2.3, 2.7, 2.0)
barplot(ctr, names.arg=date,
        col="orange",
        xlab="Date",
        ylab="Click Through Rate (%)",
        main="Click Through Rate by Day")
```

OUTPUT:



3. Build a table to show the website traffic data for each day. Label the table elements.

CODE:

```
# Dataset
website_data <- data.frame(
  Date=c("2023-01-01", "2023-01-02", "2023-01-03"),
  Page_Views=c(1500, 1600, 1400),
  Click_Through_Rate=c(2.3, 2.7, 2.0)
)

# Display Table
print(website_data)
```

OUTPUT:

```
print(website_data)
      Date Page_Views Click_Through_Rate
2023-01-01      1500             2.3
2023-01-02      1600             2.7
2023-01-03      1400             2.0
```

Set-13:

Dataset: Geographic Data

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

CODE:

```
# Dataset
city <- c("City A", "City B", "City C")

# Sample Coordinates (for visualization only)
longitude <- c(78.5, 77.6, 79.2)
latitude <- c(13.0, 12.9, 13.5)

plot(longitude, latitude,
      pch=19,
      col="blue",
      xlab="Longitude",
      ylab="Latitude",
      main="Geographic Distribution of Cities")

text(longitude, latitude, labels=city, pos=3)
```

OUTPUT:

	Date	Page_Views	Click_Through_Rate
1	2023-01-01	1500	2.3
2	2023-01-02	1600	2.7
3	2023-01-03	1400	2.0

2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

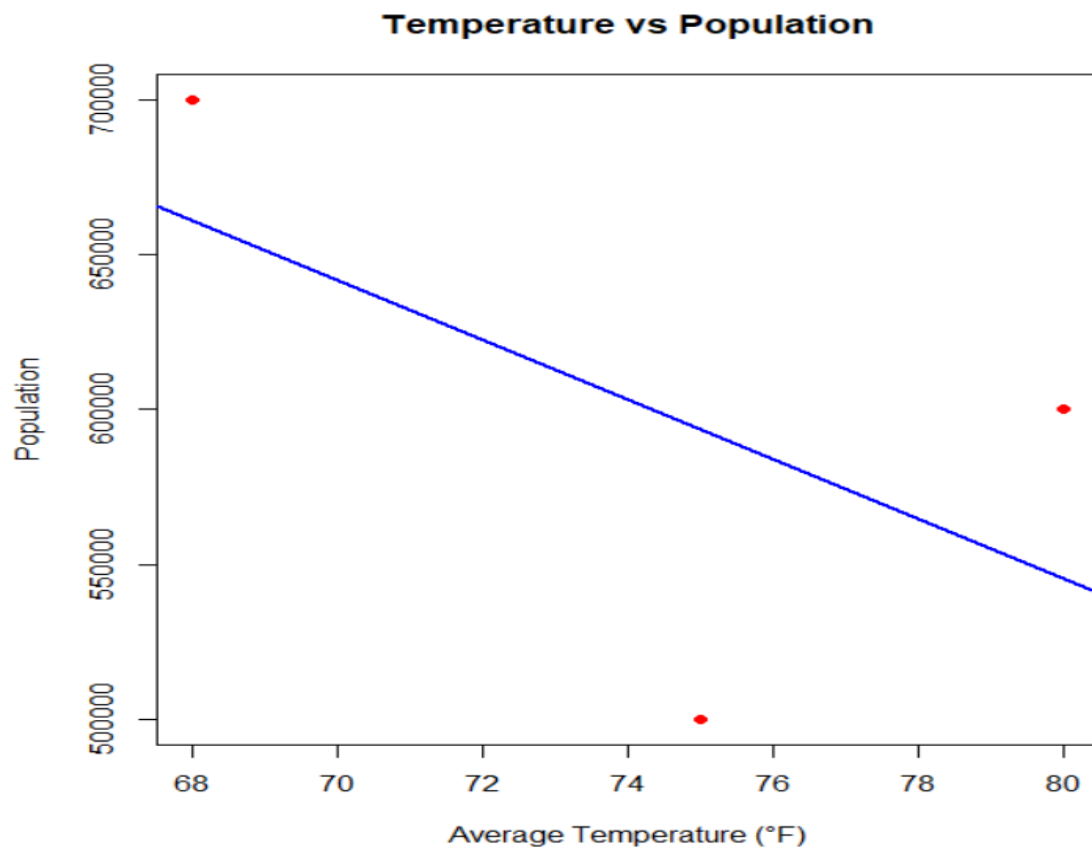
CODE:

```
# Dataset
temperature <- c(75,68,80)
population <- c(500000,700000,600000)

plot(temperature,
     population,
     pch=19,
     col="red",
     xlab="Average Temperature (°F)",
     ylab="Population",
     main="Temperature vs Population")

abline(lm(population~temperature),col="blue",lwd=2)
```

OUTPUT:



3. Build a table to display the geographic data for each city. Label the table elements.

```
CODE: # Dataset

geo_data <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  Avg_Temperature=c(75,68,80),
  Elevation=c(1000,800,1200)
)

print(geo_data)
```

OUTPUT:

	City	Population	Avg_Temperature	Elevation
1	City A	500000	75	1000
2	City B	700000	68	800
3	City C	600000	80	1200

4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Set 14:

Survey Results (R Programming)

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

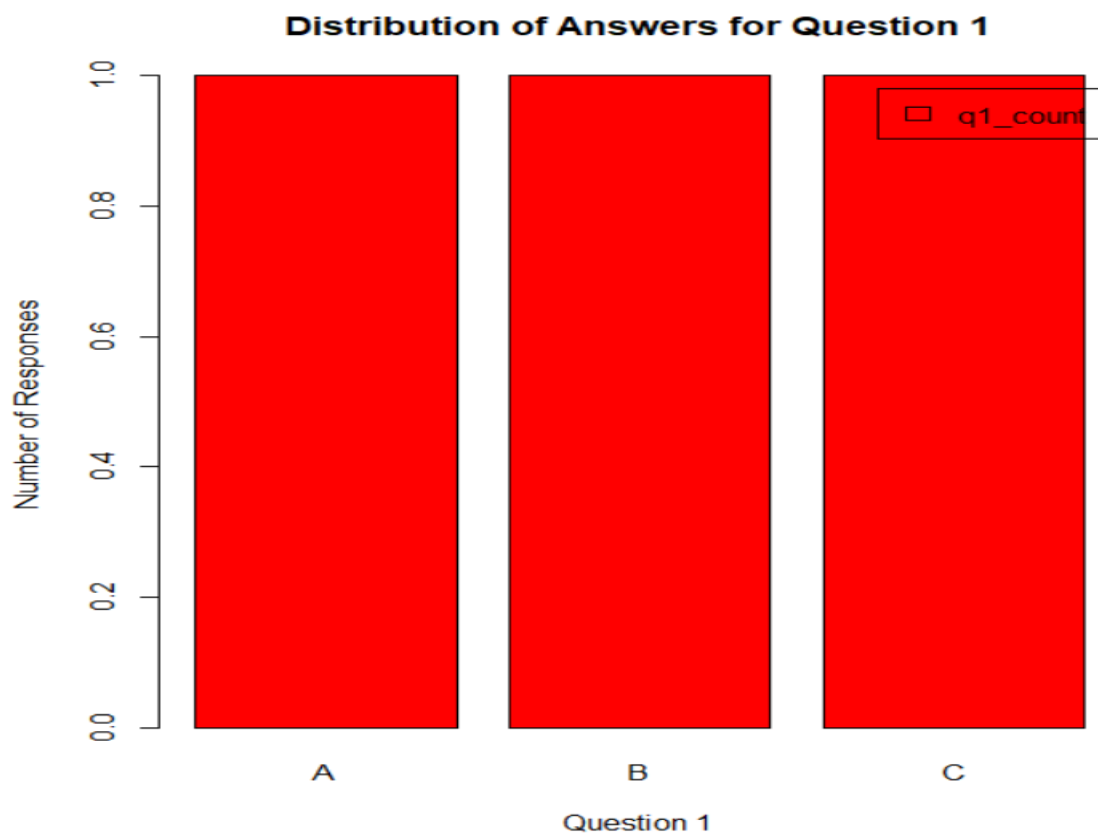
CODE:

```
# Dataset
Q1 <- c("A", "B", "C")

# Count responses
q1_count <- table(Q1)

# Stacked Bar Chart
barplot(rbind(q1_count),
        col=c("red", "green", "blue"),
        main="Distribution of Answers for Question 1",
        xlab="Question 1",
        ylab="Number of Responses",
        legend.text=rownames(rbind(q1_count)))
```

OUTPUT:



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

CODE:

```
library(fmsb)

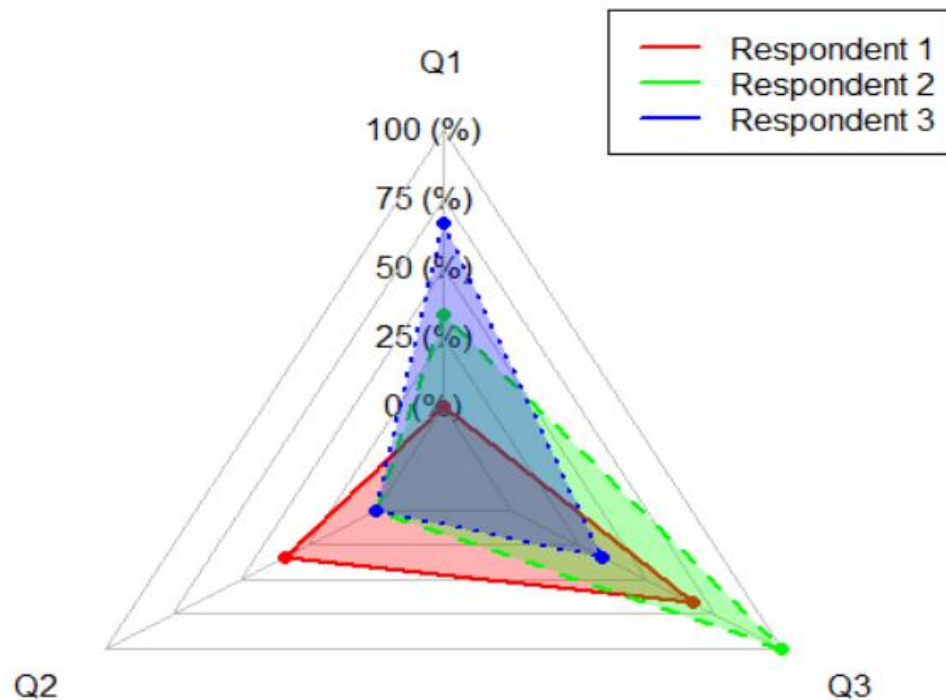
# Create data for radar chart
data <- data.frame(
  Q1 = c(4, 1, 1, 2, 3),
  Q2 = c(4, 1, 2, 1, 1),
  Q3 = c(4, 1, 3, 4, 2)
)

# Radar chart
radarchart(
  data,
  axistype = 1,
  pcol = c("red", "green", "blue"),
  pfcol = c(rgb(1,0,0,0.3),
             rgb(0,1,0,0.3),
             rgb(0,0,1,0.3)),
  plwd = 2,
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",
  title = "Survey Response Pattern"
)

legend("topright",
  legend = c("Respondent 1", "Respondent 2", "Respondent 3"),
  col = c("red", "green", "blue"),
  lty = 1,
  lwd = 2)
```

OUTPUT:

Survey Response Pattern



3. Build a table to show the survey response data for each respondent. Label the table elements.

CODE:

```
# Dataset
geo_data <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  Avg_Temperature=c(75,68,80),
  Elevation=c(1000,800,1200)
)

print(geo_data)
```

OUTPUT:

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Set 15:

Product Sales Analysis:

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

CODE:

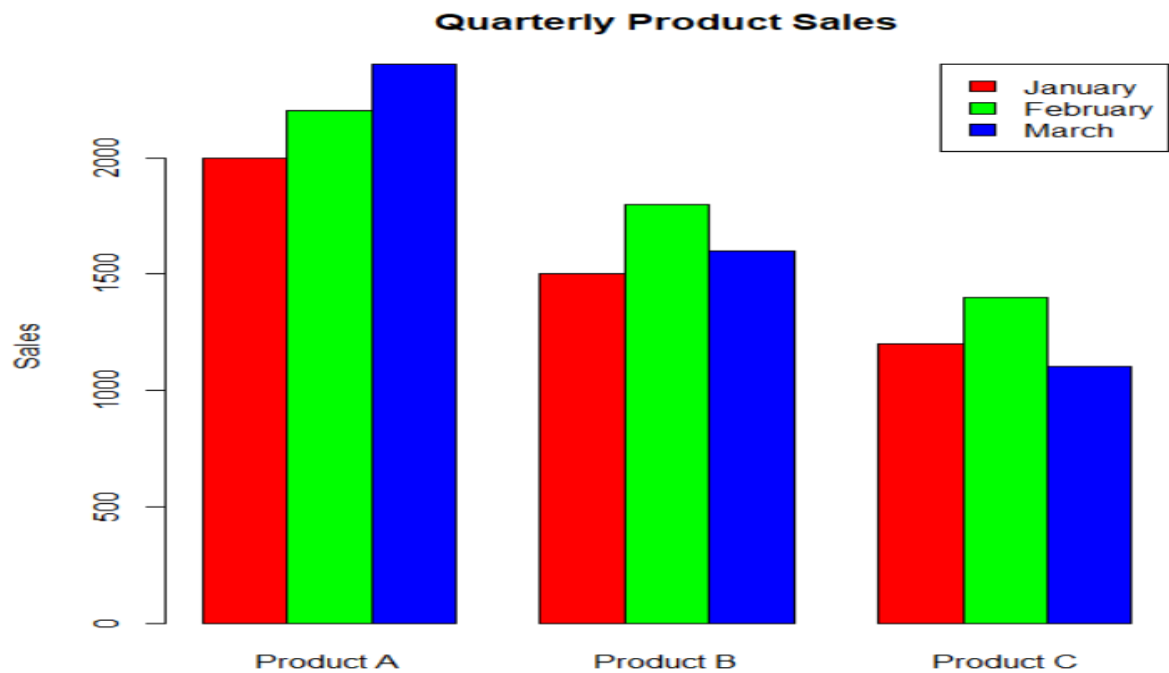
```
# Dataset
product <- c("Product A", "Product B", "Product C")

sales <- matrix(c(
  2000,1500,1200,
  2200,1800,1400,
  2400,1600,1100),
  nrow=3,
  byrow=TRUE)

barplot(sales,
        beside=TRUE,
        col=c("red", "green", "blue"),
        names.arg=product,
        xlab="Products",
        ylab="Sales",
        main="Quarterly Product Sales")

legend("topright",
       legend=c("January", "February", "March"),
       fill=c("red", "green", "blue"))
```

OUTPUT:



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
CODE: # Dataset

month <- c("January", "February", "March")

productA <- c(2000, 2200, 2400)
productB <- c(1500, 1800, 1600)
productC <- c(1200, 1400, 1100)

# Create matrix
sales <- rbind(productA, productB, productC)

# Stacked Area Chart
x <- 1:3

plot(x, cumsum(sales)[3,],
     type="n",
     xaxt="n",
     xlab="Month",
     ylab="Sales",
     main="Stacked Area Chart of Product Sales")

axis(1, at=x, labels=month)

polygon(c(x, rev(x)),
       c(rep(0,3), rev(productA)),
       col="red", border="black")
```

```

polygon(c(x, rev(x)),
       c(productA, rev(productA + productB)),
       col="green", border="black")

polygon(c(x, rev(x)),
       c(productA + productB,
         rev(productA + productB + productC)),
       col="blue", border="black")

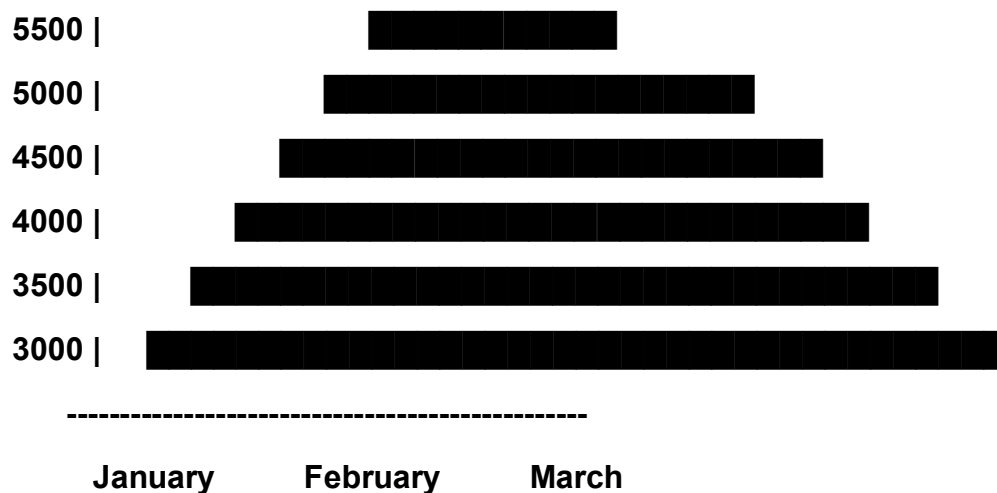
legend("topleft",
      legend=c("Product A", "Product B", "Product C"),
      fill=c("red", "green", "blue"))

```

OUTPUT:

Stacked Area Chart of Product Sales

Sales



Legend

- Red - Product A
- Green - Product B
- Blue - Product C

3. Build a table to show the monthly sales data for each product. Label the table elements.

CODE: # Dataset

```
sales_data <- data.frame(  
  Product=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100)  
)  
  
print(sales_data)
```

output:

	Product	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100